Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/08/2024
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	The Columnar to Equiaxed Transition and Dendrite Fragmentation		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	p-voorhees@northwestern.edu	**Fax:**	FY 847-491-7820
PI Organization Type:	UNIVERSITY	**Phone:**	847-491-7815
Organization Name:	Northwestern University		
PI Address 1:	Department for Materials Science and Engineering		
PI Address 2:	2220 Campus Dr		
PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	01/01/2017	**End Date:**	08/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rabenberg, Ellen	**Contact Phone:**	(256) 544-7264
Contact Email:	ellen.m.rabenberg@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2022 per NSSC information (Ed., 10/12/22). NOTE: End date changed to 7/31/2022 per P.I. (Ed., 1/10/22). NOTE: End date changed to 4/30/2022 per NSSC information (Ed., 1/4/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gulsoy, Emine Ph.D. (Northwestern University)		
Grant/Contract No.:	NNX17AD48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The shape and size of dendritic grain structures in engineering alloys is central to the resulting properties of the material. For example, equiaxed grain structures tend to provide more isotropic materials properties then materials with columnar grain morphologies. By contrast, in other applications, such as NI-base superalloys, columnar grain structures provide superior high temperature or creep properties. It is thus of critical importance to understand and predict the factors affecting the formation of grain morphology during solidification. The processes controlling the formation of equiaxed and columnar grains during solidification will be investigated using two approaches. The first is to measure the grain structures of Al-Si alloys processed by the Columnar-to-Equiaxed Transition in SOLidification Processing (CETSOL) team that is sponsored by the European Space Agency (ESA), and the second is to examine the effects of solidification on the dendrite fragmentation process. By processing these samples on the International Space Station (ISS), both sedimentation of the dendritic fragments and buoyancy driven convection can be avoided. This will permit an understanding of the physics underlying these processes to be gained, and enable a careful comparison with simulation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In nearly all casting processes, the resulting solid consists of an array of grains of different crystallographic orientation. Depending on the processing parameters, columnar or equiaxed grain growth is present, while in others there is a transition from columnar to equiaxed grains (CET). The importance of the CET lies in the clear connection between the properties of the solidified ingots and the grain structure. For example, ingots with elongated grains can have much different creep behavior than those with equiaxed grains. The grains in this case can be a single dendrite, as in the equiaxed region, or a collection of dendrites with the same crystallographic orientation. In many cases, such as in hypoeutectic alloys during cooling a mush of dendrites and liquid is formed. The last to solidify liquid forms a eutectic both between the secondary dendrite arms, but more importantly in equiaxed structures between the grains. This can lead to casting defects, such as hot tearing and intergranular cracking. Thus, the conditions under which an ingot solidifies with a columnar or equiaxed structure can have a major effect on the properties of the resulting part. Moreover, if the conditions under which the CET occurs are well understood, a properly placed CET can be used to tailor the properties of the ingot, as well as the spatial distribution of the intergranular eutectic.
Task Progress:	uccessful experiments were performed aboard the International Space Station (ISS) of the Columnar-to-Equiaxed Transition and Dendrite Fragmentation (DFM) samples. In addition, samples from the Columnar-to-Equiaxed Transition in Solidification processing (CETSOL) microgravity experiments have been analyzed and a paper is being prepared. All the experiments outlined in the DFM Science Requirements Document (SRD) were performed on the ISS. The furnace performed as expected and yielded the desired heating and cooling rates outlined in the SRD. A preliminary analysis of several of the samples shows that the microstructures obtained are consistent with planned heating and cooling rates. In particular, the samples show dendrites with secondary arms that are necessary precursors for dendrite fragmentation experiments. We have also determined that the dendrites are the proper size needed to exhibit the effects of the interplay between coarsening and solidification as envisioned when we proposed the experiment. Furthermore, the coarsening rate is very close to what we predicted. The samples with an experiment time of 20 min and 180 min were supposed to coarsen to 22 microns and 33 microns average size scale, respectively. The actual results are within 10% of the expected results. This proves that our assumption that coarsening rates would be very similar to isothermal coarsening rates, despite involving temperature change, is well-founded. Finally, the volume fraction of solid, an important parameter setting the fragmentation rate, is within 20% of the desired value. In conclusion, the samples were processed correctly, the microstructure is what we expect, and the solidification and coarsening processes occurred at the rates predicted. Thus, we expect that a thorough analysis of these samples will yield provide important insights into the mechanisms of the dendrite fragmentation process, and will yield verified models that can be incorporated into casting and additive manufacturing simulations. B. CETSOL The three-dimensional reconstructions of the CETSOL samples show a strong correlation of fragmentation with the growth direction of the dendrites. The actual cause of the difference in fragment density appears to be related to the misorientation of dendrite colonies. However, misorientation between a dendrite colony and the solidification direction may not be enough to cause fragmentation. The proximity of another dendrite colony is necessary. Dendrite misorientation during directional solidification can have a major effect on the production of fragments. The paper describing these results is: Abou-Khalil, et al. "Three-dimensional investigation of fragment distribution in Al–7wt.% Si solidified in microgravity." Acta Materialia 250 (2023): 118882. [Ed. Note: See Bibliography.]
Bibliography Type:	Description: (Last Updated: 01/17/2024)
Articles in Peer-reviewed Journals	Abou-Khalil L, Thompson Z, Reinhart G, Stan T, Sturz L, Zimmermann G, Voorhees PW, Mangelinck-Noël N, Nguyen-Thi H. "Three-dimensional investigation of fragment distribution in Al – 7 wt.% Si solidified in microgravity." Acta Materialia. 2023 May 15;250:118882. https://doi.org/10.1016/j.actamat.2023.118882 , May-2023
Articles in Peer-reviewed Journals	AArai N, Stan T, Macfarland S, Voorhees PW, Muyanja NS, Shahani AJ, Faber KT. "Coarsening of dendrites in solution-based freeze-cast ceramic systems." Acta Materialia. 2021 Aug 15;215:117039. https://doi.org/10.1016/j.actamat.2021.117039 , Aug-2021
Articles in Peer-reviewed Journals	Naviroj M, Voorhees PW, Faber KT. "Suspension-and solution-based freeze casting for porous ceramics." Journal of Materials Research. 2017 Sep;32(17):3372-82. http://dx.doi.org/10.1557/jmr.2017.133 , Sep-2017